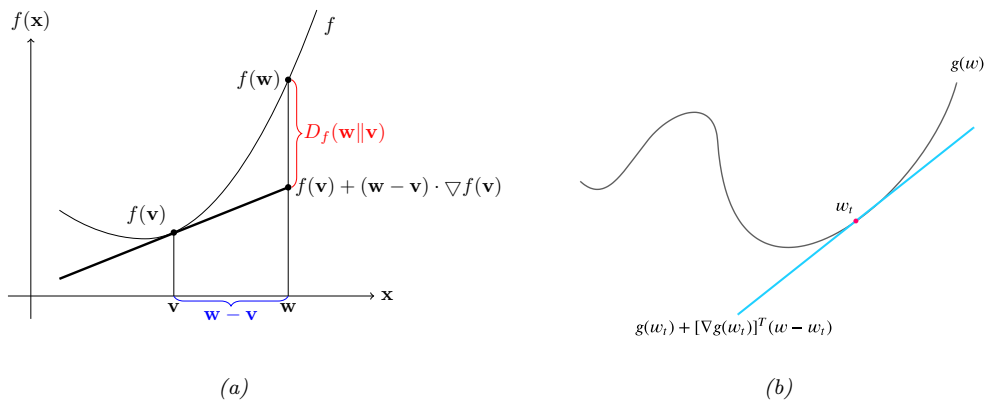

Figure 5.3: (a) Illustration of Bregman divergence. (b) A locally linear approximation to a non-convex function.

- If $f(\boldsymbol{w}) = ||\boldsymbol{w}||^2$, then $B_f(\boldsymbol{w}||\boldsymbol{v}) = ||\boldsymbol{w} - \boldsymbol{v}||^2$ is the squared Euclidean distance.
- If $f(\boldsymbol{w}) = \boldsymbol{w}^\mathsf{T}\mathbf{Q}\boldsymbol{w}$, then $B_f(\boldsymbol{w}||\boldsymbol{v})$ is the squared Mahalanobis distance.
- If $\boldsymbol{w}$ are the natural parameters of an exponential family distribution, and $f(\boldsymbol{w}) = \log Z(\boldsymbol{w})$ is the log normalizer, then the Bregman divergence is the same as the Kullback-Leibler divergence, as we show in Section 5.1.10.1.

5.1.10.1 KL is a Bregman divergence

Recall that the log partition function $A(\boldsymbol{\eta})$ is a convex function. We can therefore use it to define the Bregman divergence (Section 5.1.10) between the two distributions, p and q, as follows:

$$B_f(\boldsymbol{\eta}_q||\boldsymbol{\eta}_p) = A(\boldsymbol{\eta}_q) - A(\boldsymbol{\eta}_p) - (\boldsymbol{\eta}_q - \boldsymbol{\eta}_p)^\mathsf{T}\nabla_{\boldsymbol{\eta}_p} A(\boldsymbol{\eta}_p) \tag{5.86}$$

$$= A(\boldsymbol{\eta}_q) - A(\boldsymbol{\eta}_p) - (\boldsymbol{\eta}_q - \boldsymbol{\eta}_p)^\mathsf{T}\mathbb{E}_p\left[\mathcal{T}(\boldsymbol{x})\right] \tag{5.87}$$

$$= D_{\mathbb{KL}}\left(p \parallel q\right) \tag{5.88}$$

where we exploited the fact that the gradient of the log partition function computes the expected sufficient statistics as shown in Section 2.4.3.

In fact, the KL divergence is the only divergence that is both a Bregman divergence and an f-divergence (Section 2.7.1) [Ama09].

5.2 Entropy

In this section, we discuss the **entropy** of a distribution p, which is just a shifted and scaled version of the KL divergence between the probability distribution and the uniform distribution, as we will see.

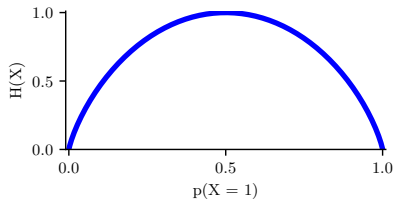

Figure 5.4: Entropy of a Bernoulli random variable as a function of θ. The maximum entropy is $\log_2 2 = 1$. Generated by bernoulli_entropy_fig.ipynb.

5.2.1 Definition

The entropy of a discrete random variable X with distribution p over K states is defined by

$$\mathbb{H}(X) \triangleq -\sum_{k=1}^{K} p(X=k) \log p(X=k) = -\mathbb{E}_X [\log p(X)] \tag{5.89}$$

We can use logarithms to any base, but we commonly use log base 2, in which case the units are called bits, or log base e, in which case the units are called nats, as we explained in Section 5.1.3.1.

The entropy is equivalent to a constant minus the KL divergence from the uniform distribution:

$$\mathbb{H}(X) = \log K - D_{\mathbb{KL}}(p(X) \parallel u(X)) \tag{5.90}$$

$$D_{\mathbb{KL}}(p(X) \parallel u(X)) = \sum_{k=1}^{K} p(X=k) \log \frac{p(X=k)}{\frac{1}{K}} \tag{5.91}$$

$$= \log K + \sum_{k=1}^{K} p(X=k) \log p(X=k) \tag{5.92}$$

If p is uniform, the KL is zero, and we see that the entropy achieves its maximal value of $\log K$.

For the special case of binary random variables, $X \in \{0, 1\}$, we can write $p(X = 1) = \theta$ and $p(X = 0) = 1 - \theta$. Hence the entropy becomes

$$\mathbb{H}(X) = -[p(X=1) \log p(X=1) + p(X=0) \log p(X=0)] \tag{5.93}$$

$$= -[\theta \log \theta + (1-\theta) \log(1-\theta)] \tag{5.94}$$

This is called the **binary entropy function**, and is also written $\mathbb{H}(\theta)$. We plot this in Figure 5.4. We see that the maximum value of 1 bit occurs when the distribution is uniform, $\theta = 0.5$. A fair coin requires a single yes/no question to determine its state.

5.2.2 Differential entropy for continuous random variables

If X is a continuous random variable with pdf $p(x)$, we define the **differential entropy** as

$$h(X) \triangleq -\int_{\mathcal{X}} dx\, p(x) \log p(x) \tag{5.95}$$

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assuming this integral exists.

For example, one can show that the entropy of a d-dimensional Gaussian is

$$h(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})) = \frac{1}{2}\log|2\pi e\boldsymbol{\Sigma}| = \frac{1}{2}\log[(2\pi e)^d|\boldsymbol{\Sigma}|] = \frac{d}{2} + \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|\boldsymbol{\Sigma}| \tag{5.96}$$

In the 1d case, this becomes

$$h(\mathcal{N}(\mu, \sigma^2)) = \frac{1}{2}\log\left[2\pi e\sigma^2\right] \tag{5.97}$$

Note that, unlike the discrete case, *differential entropy can be negative*. This is because pdf's can be bigger than 1. For example, suppose $X \sim U(0, a)$. Then

$$h(X) = -\int_0^a dx\, \frac{1}{a}\log\frac{1}{a} = \log a \tag{5.98}$$

If we set $a = 1/8$, we have $h(X) = \log_2(1/8) = -3$ bits.

One way to understand differential entropy is to realize that all real-valued quantities can only be represented to finite precision. It can be shown [CT91, p228] that the entropy of an n-bit quantization of a continuous random variable X is approximately $h(X) + n$. For example, suppose $X \sim U(0, \frac{1}{8})$. Then in a binary representation of X, the first 3 bits to the right of the binary point must be 0 (since the number is $\leq 1/8$). So to describe X to n bits of accuracy only requires $n - 3$ bits, which agrees with $h(X) = -3$ calculated above.

The continuous entropy also lacks the reparameterization independence of KL divergence (Section 5.1.2.3). In particular, if we transform our random variable $y = f(x)$, the entropy transforms. To see this, note that the change of variables tells us that

$$p(y)\,dy = p(x)\,dx \implies p(y) = p(x)\left|\frac{dy}{dx}\right|^{-1}, \tag{5.99}$$

Thus the continuous entropy transforms as follows:

$$h(X) = -\int dx\, p(x)\log p(x) = h(Y) - \int dy\, p(y)\log\left|\frac{dy}{dx}\right|. \tag{5.100}$$

We pick up a factor in the continuous entropy of the log of the determinant of the Jacobian of the transformation. This changes the value for the continuous entropy even for simply rescaling the random variable such as when we change units. For example in Figure 5.5 we show the distribution of adult human heights (it is bimodal because while both male and female heights are normally distributed, they differ noticeably). The continous entropy of this distribution depends on the units it is measured in. If measured in feet, the continuous entropy is 0.43 bits. Intuitively this is because human heights mostly span less than a foot. If measured in centimeters it is instead 5.4 bits. There are 30.48 centimeters in a foot, $\log_2 30.48 = 4.9$ explaining the difference. If we measured the continuous entropy of the same distribution measured in meters we would obtain -1.3 bits!

5.2.3 Typical sets

The **typical set** of a probability distribution is the set whose elements have an information content that is close to that of the expected information content from random samples from the distribution.